

Random Variables

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Counting heads

Problem

Flip a fair coin twice independently and record the number of heads.
What numbers can be recorded, and how likely is each number?

Recording a number

Flip a fair coin twice independently. With H for heads and T for tails, the four equally likely outcomes are

$$\Omega = \{HH, HT, TH, TT\}, \quad p(\omega) = 1/4.$$

Suppose we record only the number of heads:

Outcome ω	HH	HT	TH	TT
Number of heads	2	1	1	0

Different outcomes may produce the same number. The number of heads is an example of a **random variable**.

Random variables

A **random variable** assigns a real number to each outcome:

$$X : \Omega \longrightarrow \mathbb{R}$$

For example, let X count heads in two coin flips:

$$X(HH) = 2, \quad X(HT) = X(TH) = 1, \quad X(TT) = 0.$$

A **random variable** is called “random” because its resulting value depends on the outcome of a random event or process, even though the mathematical rule itself is fixed.

A random variable is **discrete** if its possible values form a finite or countably infinite set.

Why use random variables?

In mathematics, naming an object lets us refer to it repeatedly and study its properties.

A familiar example. We can calculate $2 + 1$ or $10 + 1$ without defining a function. With $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x + 1$, we can write $f(2)$ and $f(10)$ and study the same rule for any input.

The same idea in probability. We can list the coin-flip outcomes and count heads directly. Defining

$$X(\omega) = \text{the number of heads in outcome } \omega$$

gives this measurement a name. We can then study how likely each value of X is, its average, and how much it varies.

A random variable lets us organize many questions about **the same quantity of interest**.

Three measurements of two dice

Problem

Roll a fair red die and a fair blue die independently. From the same roll, record:

- ▶ the sum of the two results;
- ▶ the larger result;
- ▶ whether the results match.

How can we express these three measurements as random variables?

Three measurements: solution

Sample space and probabilities. Record the red and blue results as (i, j) :

$$\Omega = \{(i, j) : i, j \in \{1, \dots, 6\}\}, \quad p((i, j)) = 1/36.$$

Random variables. Define three functions on the same Ω :

$$X(i, j) = i + j, \quad Y(i, j) = \max(i, j), \quad Z(i, j) = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases}$$

For outcome $(2, 5)$, we get $X = 7$, $Y = 5$, and $Z = 0$. For outcome $(4, 4)$, we get $X = 8$, $Y = 4$, and $Z = 1$.

The outcome is a pair. Each random variable records a different numerical feature of that pair.

Events defined by a random variable

The notation $\{X = x\}$ abbreviates the event

$$\{\omega \in \Omega : X(\omega) = x\}.$$

For the number of heads,

$$\{X = 1\} = \{HT, TH\}, \quad P(X = 1) = \frac{2}{4} = \frac{1}{2}.$$

Likewise,

$$\{X \leq 1\} = \{TT, HT, TH\}, \quad P(X \leq 1) = \frac{3}{4}.$$

Probabilities are still assigned to *events*; X gives us a convenient way to describe those events.

Probability mass functions

The **probability mass function (PMF)** of a discrete random variable X is

$$p_X(x) = P(X = x).$$

Let S_X be the set of its possible values. Then

$$p_X(x) \geq 0, \quad \sum_{x \in S_X} p_X(x) = 1.$$

For any set B of real numbers,

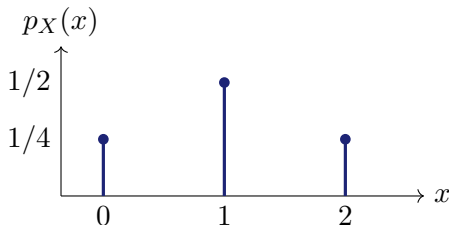
$$P(X \in B) = \sum_{x \in B \cap S_X} p_X(x).$$

The PMF describes how probability is distributed among the values of X .

The distribution of the number of heads

For two independent fair flips,

$$p_X(0) = \frac{1}{4}, \quad p_X(1) = \frac{1}{2}, \quad p_X(2) = \frac{1}{4}.$$



There are three possible values, but they are not equally likely. Two different outcomes give the value 1.

The larger die result

Problem

Roll two fair dice independently and record the larger result. What is its probability distribution?

Before calculating, which values can the larger result take? Are these values equally likely?

The larger die result: solution

Use $\Omega = \{1, \dots, 6\}^2$, with $p((i, j)) = 1/36$, and define $Y(i, j) = \max(i, j)$. The notation $\{1, \dots, 6\}^2$ means all ordered pairs of die results. For $k \in \{1, \dots, 6\}$, the target event is

$$\{Y = k\} = \{(i, j) \in \Omega : \max(i, j) = k\}.$$

Fix the first die at k . The second can be $1, \dots, k$: k pairs. Fix the second die at k . The first can be $1, \dots, k$: k pairs. The pair (k, k) appears in both counts, so subtract it once:

$$p_Y(k) = \frac{k + k - 1}{36} = \frac{2k - 1}{36}.$$

k	1	2	3	4	5	6
$36p_Y(k)$	1	3	5	7	9	11

The probabilities sum to $(1 + 3 + 5 + 7 + 9 + 11)/36 = 1$.

Cumulative distribution functions

The **cumulative distribution function (CDF)** of X is

$$F_X(t) = P(X \leq t), \quad t \in \mathbb{R}.$$

For a discrete random variable, add all masses at values at most t :

$$F_X(t) = \sum_{x \leq t} p_X(x).$$

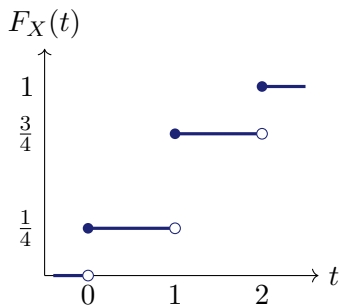
For our two-coin example,

$$F_X(0) = \frac{1}{4}, \quad F_X(0.5) = \frac{1}{4}, \quad F_X(1) = \frac{3}{4}, \quad F_X(2) = 1.$$

The input t may be any real number, not just a possible value of X .

A discrete CDF

$$F_X(t) = \begin{cases} 0 & t < 0, \\ 1/4 & 0 \leq t < 1, \\ 3/4 & 1 \leq t < 2, \\ 1 & t \geq 2. \end{cases}$$



A filled dot shows the value *at* the endpoint. Because the definition uses $\leq$, the mass at t is already included in $F_X(t)$.

Reading probabilities from a CDF

For $a < b$, the event $\{X \leq b\}$ is the disjoint union of $\{X \leq a\}$ and $\{a < X \leq b\}$. Hence

$$P(a < X \leq b) = F_X(b) - F_X(a).$$

For the two-coin example,

$$P(0 < X \leq 1) = F_X(1) - F_X(0) = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}.$$

Be careful with endpoints:

$$P(0 \leq X \leq 1) = \frac{3}{4} \neq P(0 < X \leq 1).$$

The difference is the mass $P(X = 0) = 1/4$.

Properties of a CDF

Every CDF satisfies:

- ▶ $0 \leq F_X(t) \leq 1$.
- ▶ F_X is nondecreasing: if $s \leq t$, then $\{X \leq s\} \subseteq \{X \leq t\}$.
- ▶ $F_X(t) \rightarrow 0$ as $t \rightarrow -\infty$, and $F_X(t) \rightarrow 1$ as $t \rightarrow \infty$.

At a possible value x of a discrete random variable, the jump in the CDF equals $P(X = x)$.

The CDF definition applies to both discrete and continuous models.

Continuous models

A count, such as the number of heads, has discrete values. A measurement, such as a waiting time, is often modeled continuously. For a random variable with a **probability density function (PDF)** f_X , probabilities are areas:

$$P(a < X \leq b) = \int_a^b f_X(t) dt.$$

The density satisfies

$$f_X(t) \geq 0, \quad \int_{-\infty}^{\infty} f_X(t) dt = 1.$$

The discrete PMF gives *probability at a point*. A PDF gives *density*, not probability at a point.

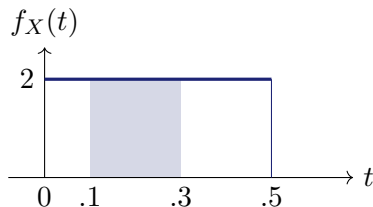
A continuous measurement

Problem

A waiting time is modeled as uniform between zero and half a minute.
What is the probability of waiting between 0.1 and 0.3 minutes?

A uniform density

Model. Use $\Omega = [0, 1/2]$ and $X(\omega) = \omega$. Probabilities come from density $f_X(t) = 2$ on this interval, and 0 elsewhere, not from assigning positive masses to points.



$$A = \{\omega \in \Omega : 0.1 < \omega \leq 0.3\}, \quad P(A) = 2(0.3 - 0.1) = 0.4.$$

The height is $2 > 1$, but the total area is $2(1/2) = 1$. For this model, $P(X = t) = 0$ for each individual point t .

PMF, PDF, and CDF

	Discrete	Has a density
Function	PMF $p_X(x)$	PDF $f_X(x)$
Probability	Sum of masses	Area under density
CDF	$\sum_{x \leq t} p_X(x)$	$\int_{-\infty}^t f_X(x) dx$

In both cases, $F_X(t) = P(X \leq t)$. If the density f_X is continuous at t , the fundamental theorem of calculus gives $F'_X(t) = f_X(t)$.

Not every distribution has a PDF. We now return to discrete random variables.

Bernoulli random variables

A trial has two possible results, called success and failure. Let the probability of success be p , with $0 \leq p \leq 1$. Define

$$X = \begin{cases} 1 & \text{if the trial succeeds,} \\ 0 & \text{otherwise.} \end{cases}$$

Then

$$P(X = 1) = p, \quad P(X = 0) = 1 - p.$$

We say that X has a **Bernoulli distribution**, and write $X \sim \text{Bernoulli}(p)$. For example, X may indicate whether a randomly selected user clicks an ad. “Success” is simply the outcome we have chosen to count.

Repeated independent trials

Perform n independent trials, each with success probability p . Here independence means that the probability of any specified sequence of results is the product of the individual probabilities. For example, with three trials,

$$P(\text{success, failure, success}) = p(1 - p)p.$$

Let X be the total number of successes. Then X takes values $0, 1, \dots, n$.

To find $P(X = k)$, we must account for *all* sequences with exactly k successes.

The binomial distribution

First assume $0 < p < 1$. Each sequence with k successes and $n - k$ failures has probability

$$p^k(1 - p)^{n-k}.$$

Choose which k of the n trial positions are successes. There are $\binom{n}{k}$ such sequences, and they are disjoint outcomes.

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, \dots, n.$$

We write $X \sim \text{Binomial}(n, p)$.

The model requires a fixed number of independent trials with the same success probability. At $p = 0$ or $p = 1$, the count is constant.

Counting clicks

Problem

Five users independently see an advertisement. Each clicks with probability 0.2.

What is the probability of exactly two clicks? What is the probability of at least one click?

Counting clicks: solution

Write $c_i = 1$ if user i clicks and $c_i = 0$ otherwise.

$$\Omega = \{0, 1\}^5, \quad X(c_1, \dots, c_5) = \sum_{i=1}^5 c_i.$$

For an outcome with k clicks, independence gives $p(\omega) = (0.2)^k(0.8)^{5-k}$. The target events are

$$A = \{\omega : X(\omega) = 2\}, \quad B = \{\omega : X(\omega) \geq 1\}.$$

Since $X \sim \text{Binomial}(5, 0.2)$,

$$P(A) = \binom{5}{2} (0.2)^2 (0.8)^3 = 0.2048, \quad P(B) = 1 - (0.8)^5 = 0.67232.$$

Real users need not have identical or independent behavior. These are modeling assumptions in this example.

Waiting for a success

Problem

A data transmission succeeds with probability 0.2 on each attempt. Attempts are independent, and transmission is repeated until it succeeds.

What is the probability that the first success occurs on the third attempt? What is the probability of needing more than three attempts?

Waiting for a success: solution

Write F for failure and S for success. An outcome is the sequence up to the first success:

$$\Omega = \{S, FS, FFS, FFFS, \dots\}.$$

Let $T(\omega)$ be the number of attempts in that sequence. For a sequence of $k - 1$ failures followed by a success,

$$p(F^{k-1}S) = (0.8)^{k-1}(0.2), \quad k = 1, 2, \dots$$

Here $F^{k-1}S$ means $k - 1$ copies of F followed by S . The target events and probabilities are

$$A = \{T = 3\} = \{FFS\}, \quad P(A) = (0.8)^2(0.2) = 0.128,$$

$$B = \{T > 3\} = \{FFFS, FFFFS, \dots\}, \quad P(B) = (0.8)^3 = 0.512.$$

Needing more than three attempts means the first three all fail.

The geometric distribution

For independent trials with the same success probability $0 < p < 1$, let T count trials up to and including the first success. To have $T = k$, the first $k - 1$ trials must fail and trial k must succeed:

$$P(T = k) = (1 - p)^{k-1}p, \quad k = 1, 2, \dots$$

We write $T \sim \text{Geometric}(p)$. The probabilities sum to one by the geometric series formula:

$$\sum_{k=1}^{\infty} (1 - p)^{k-1}p = \frac{p}{1 - (1 - p)} = 1.$$

The probability of failing forever is zero, since $(1 - p)^m \rightarrow 0$. Our convention counts the successful trial; it does not count only failures.

Geometric tail probabilities

For $T \sim \text{Geometric}(p)$ and integer $m \geq 0$,

$$\{T > m\} = \{\text{the first } m \text{ trials all fail}\}.$$

Consequently,

$$P(T > m) = (1 - p)^m, \quad P(T \leq m) = 1 - (1 - p)^m.$$

With success probability 0.2,

$$P(T \leq 3) = 1 - (0.8)^3 = 0.488.$$

Compare the questions: $T = 3$ means success for the first time *on* attempt three; $T \leq 3$ means success *by* attempt three.

Binomial or geometric?

Both models use independent trials with the same success probability p . They record different quantities.

Binomial: fix the number of trials n , then count successes.

$$X \in \{0, \dots, n\}, \quad P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}.$$

Geometric: stop at the first success, then count trials.

$$T \in \{1, 2, \dots\}, \quad P(T = k) = (1 - p)^{k-1} p.$$

Choose the random variable from the question before choosing a formula.

Discrete distributions

Distribution	What it records	Values
Bernoulli	Success of one trial	0, 1
Binomial	Successes in n trials	$0, \dots, n$
Geometric	Trials until first success	$1, 2, \dots$

For each model, the PMF describes probabilities at individual values, and the CDF accumulates those probabilities.

A distribution describes the random variable's values and probabilities, not the full details of the original experiment.

Requests over a time interval

A server receives $\lambda > 0$ requests on average over $t > 0$ minutes. How can we model the number of requests in $[0, t]$?

Assume a constant arrival rate, independent counts in disjoint intervals, and negligible multiple arrivals in very short intervals.

- ▶ Divide $[0, t]$ into n equal intervals of length t/n .
- ▶ Approximate each interval by an independent trial: one request with probability λ/n , otherwise none.
- ▶ Let X_n count requests in this approximation. For $n > \lambda$,

$$X_n \sim \text{Binomial}(n, \lambda/n).$$

The choice λ/n keeps the average total at $n(\lambda/n) = \lambda$.

As $n \rightarrow \infty$, the intervals get shorter. **The total observation time remains t .**

A limit of binomial distributions

In the approximation above, λ stays fixed while n increases:

$$X_n \sim \text{Binomial}(n, \lambda/n).$$

For a fixed integer $k \geq 0$ and $n > \max\{\lambda, k\}$,

$$\begin{aligned} P(X_n = k) &= \binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \\ &= \underbrace{\frac{n!}{(n-k)! n^k}}_{\rightarrow 1} \frac{\lambda^k}{k!} \underbrace{\left(1 - \frac{\lambda}{n}\right)^n}_{\rightarrow e^{-\lambda}} \underbrace{\left(1 - \frac{\lambda}{n}\right)^{-k}}_{\rightarrow 1}. \end{aligned}$$

Here e is the base of the natural logarithm. Thus, as $n \rightarrow \infty$,

$$\boxed{P(X_n = k) \rightarrow e^{-\lambda} \frac{\lambda^k}{k!} .}$$

This limit gives the **Poisson distribution**.

The Poisson distribution

A random variable N has a **Poisson distribution** with parameter $\lambda > 0$, written $N \sim \text{Poisson}(\lambda)$, if its PMF is

$$P(N = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

Here $0! = 1$. The possible values are counts: $0, 1, 2, \dots$. The parameter λ is the average count in the chosen interval, not the probability of an arrival. It can be greater than 1.

Although time is continuous, the **number of arrivals is discrete**. The limiting binomial model explains why this distribution is useful for counting arrivals under the preceding assumptions.

Counting requests

Problem. Under a Poisson model, a server receives an average of two requests per minute. What is the probability of exactly three requests in a minute? Of none?

Solution. Model the recorded count directly:

$$\Omega = \{0, 1, 2, \dots\}, \quad N(\omega) = \omega, \quad p(k) = e^{-2} \frac{2^k}{k!}.$$

The target events are $A = \{3\}$ and $B = \{0\}$.

$$P(A) = e^{-2} \frac{2^3}{3!} \approx 0.1804, \quad P(B) = e^{-2} \approx 0.1353.$$

The Poisson model is an assumption here. Knowing the average alone does not determine these probabilities.